THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapters 3 & 4: **Pair of Linear Equations in Two Variables, Quadratic Equations** | Class 10 Mathematics | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** no solution	$\frac{3}{6}=\frac{4}{8}\neq\frac{5}{7}$
2	**(b)** $(3,1)$	Adding: $3x=9$
3	**(a)** 6	$\frac{2}{4}=\frac{3}{k}=\frac{5}{10}$
4	**(a)** Incorrect — they intersect at $(2,2)$	One is vertical, the other horizontal
5	**(a)** Student A is correct	$3+2=5,\ 1+4=5$
6	**(b)** $3,4$	$(x-3)(x-4)$
7	**(c)** -4	$16-20$
8	**(c)** 9	$36-4k=0$
9	**(c)** Incorrect — $x=0$ is also a root	$x(x-4)=0$
10	**(c)** $x^2-5x+6=0$	$D=25-24=1>0$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$2x=11-3y \Rightarrow 11-7y=-24 \Rightarrow y=5,\ x=-2$	1+1
12	$\frac{k}{6}\neq\frac{-1}{-2} \Rightarrow k\neq 3$	2
13	$(\sqrt{3}x-\sqrt{2})^2=0 \Rightarrow x=\sqrt{\frac{2}{3}}$ (twice)	1+1
14	$D=9-40=-31<0 \Rightarrow$ no real roots	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$x+y=12,\ 2x=y+3 \Rightarrow x=5,\ y=7 \Rightarrow \frac{5}{7}$	1+1+1
16	$x^2-3x-1=0;\ D=13 \Rightarrow x=\frac{3\pm\sqrt{13}}{2}$	1+1+1
17	Points $(0,-4),(2,0)$ and $(0,-1),(-1,0)$; lines meet at $(1,-2)$	2+1
	OR: $3B=2A+2B \Rightarrow B=2A;\ A+2A+6A=180^\circ \Rightarrow A=20^\circ,\ B=40^\circ,\ C=120^\circ$	1+1+1
18	$x+12=\frac{160}{x} \Rightarrow x^2+12x-160=0 \Rightarrow (x+20)(x-8)=0 \Rightarrow 8$	1+1+1
	OR: $7x^2-50x+7=0 \Rightarrow (7x-1)(x-7)=0 \Rightarrow 7$ or $\frac{1}{7}$	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $2s+3k=65,\ 3s+2k=60$ (ii) $s+k=25,\ k-s=5 \Rightarrow s=₹10,\ k=₹15$ (iii) $50+60=₹110$	1+2+2
	OR: $2x+y=160,\ 4x+2y=300$ (1). The lines are parallel ($\frac{2}{4}=\frac{1}{2}\neq\frac{160}{300}$) (3) $\Rightarrow$ no solution: prices must have changed (1)	1+3+1
20	(i) $(20+2w)(10+2w)-200=216 \Rightarrow w^2+15w-54=0$ (ii) $(w+18)(w-3)=0 \Rightarrow$ 3 m (iii) ₹10,800	1+2+2
	OR: $\frac{1500}{x}-\frac{1500}{x+100}=\frac{1}{2} \Rightarrow x^2+100x-300000=0 \Rightarrow (x+600)(x-500)=0 \Rightarrow$ 500 km/h. Verify: $3-2.5=0.5$ h ✓	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapters 3 & 4: **Pair of Linear Equations in Two Variables, Quadratic Equations** | Class 10 Mathematics | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** infinitely many solutions	Coincident lines
2	**(d)** $(6, 4)$	$2x = 12$
3	**(d)** 2	$\frac{3}{2k-1} = \frac{1}{k-1} \Rightarrow k = 2$; then $\frac{1}{5} \neq 1$
4	**(c)** Incorrect — it has 0, 1 or infinitely many solutions	Two distinct lines meet at most once
5	**(a)** Student A is correct	Put $x = 0$; B found the x-intercept
6	**(d)** $0, -5$	$x(x + 5) = 0$
7	**(c)** −8	$4 + 2k + 12 = 0$
8	**(c)** $x^2 - 8x + 16 = 0$	$(x - 4)^2$
9	**(a)** Incorrect — it has two equal real roots	$x = -\frac{b}{2a}$ (repeated)
10	**(b)** $x^2 + 6x - 40 = 0$	$(x + 3)^2 = 49$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$9x - 15y = 12$ and $9x - 2y = 7 \Rightarrow -13y = 5,\ y = -\frac{5}{13},\ x = \frac{9}{13}$	1+1
12	$\frac{2}{4} = \frac{-3}{-6} \neq \frac{8}{9} \Rightarrow$ parallel ⇒ **inconsistent**	1+1
13	$16x^2 - 8x + 1 = 0 \Rightarrow (4x - 1)^2 = 0 \Rightarrow \frac{1}{4}, \frac{1}{4}$	1+1
14	$16 - 4p = 0 \Rightarrow p = 4$	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$10x + y = 4(x + y) \Rightarrow y = 2x$; $9x - 9y = -18 \Rightarrow y - x = 2 \Rightarrow x = 2, y = 4 \Rightarrow 24$	1+1+1
16	$-2 = 3x^2 - 6x \Rightarrow 3x^2 - 6x + 2 = 0 \Rightarrow x = \frac{3 \pm \sqrt{3}}{3}$	1+1+1
17	$b + g = 10,\ g - b = 4 \Rightarrow$ girls 7, boys 3	1+1+1
	OR: $A + C = 180 \Rightarrow y - x = 40$; $B + D = 180 \Rightarrow 3y - 7x = 180 \Rightarrow x = -15,\ y = 25$; $\angle A = 120^\circ, \angle B = 70^\circ, \angle C = 60^\circ, \angle D = 110^\circ$	1+1+1
18	$x^2 + (x + 5)^2 = 625 \Rightarrow x^2 + 5x - 300 = 0 \Rightarrow (x + 20)(x - 15) = 0 \Rightarrow$ 15 cm and 20 cm	1+1+1
	OR: $36 + 8k \geq 0 \Rightarrow k \geq -\frac{9}{2},\ k \neq 0$	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $y = \frac{x}{2}$ i.e. $x - 2y = 0$; $3x + 4y = 20$ (ii) lines meet at $(4, 2)$ (iii) 4 rides, 2 games	2+2+1
	OR: (i) $a + c = 100,\ 5a + 3c = 420 \Rightarrow a = 60,\ c = 40$ (ii) $5a + 3(100 - a) = 460 \Rightarrow a = 80$	3+2
20	(i) $\frac{80}{x} - \frac{80}{x+4} = 1 \Rightarrow x^2 + 4x - 320 = 0$ (ii) $(x + 20)(x - 16) = 0 \Rightarrow$ 16 (iii) ₹5	2+2+1
	OR: $xy = 24,\ x - y = 2 \Rightarrow y^2 + 2y - 24 = 0 \Rightarrow y = 4,\ x = 6 \Rightarrow 64$ (check $64 - 18 = 46$ ✓)	2+2+1

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ANSWER KEY & MARKING SCHEME SET C

Chapters 3 & 4: **Pair of Linear Equations in Two Variables, Quadratic Equations** | Class 10 Mathematics | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** parallel to the x-axis	All points have $y = -3$
2	**(d)** 3	$21a = 63$
3	**(d)** infinitely many solutions	All ratios equal $\frac{5}{3}$
4	**(c)** Incorrect — the lines intersect	Unique solution
5	**(b)** Student A is correct	$5 - 3 = 2$
6	**(d)** $\frac{3}{2}$	$(2x-3)(x+2)$
7	**(d)** 32	$8 + 24$
8	**(b)** real and equal	$D = 144 - 144 = 0$
9	**(c)** Incorrect — $D = 9 - 20 < 0$	Nature depends on D
10	**(a)** $x^2 + 2x - 63 = 0$	$x(x+2) = 63$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$3x + 4y = -6,\ 3x - y = 9 \Rightarrow y = -3,\ x = 2$	1+1
12	(i) $4x + 6y - 7 = 0$ (ii) $4x + 6y - 16 = 0$ (any correct)	1+1
13	$(3x-2)(2x+1) = 0 \Rightarrow \frac{2}{3}, -\frac{1}{2}$	1+1
14	$4(k+1)^2 - 4k^2 = 0 \Rightarrow 2k + 1 = 0 \Rightarrow k = -\frac{1}{2}$	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$m = 3s + 12,\ m = 9s - 24 \Rightarrow s = 6,\ m = 30$ years	1+1+1
16	$x^2 + (x+1)^2 = 421 \Rightarrow x^2 + x - 210 = 0 \Rightarrow (x+15)(x-14) = 0 \Rightarrow$ 14, 15	1+1+1
17	Meet at $(2, 2)$; with x-axis at $(0,0), (4,0)$; area $= \frac{1}{2} \times 4 \times 2 = 4$ sq units	2+1
	OR: $t + c = 1500,\ t - c = 500 \Rightarrow$ table ₹1000, chair ₹500	1+1+1
18	$D = 16b^2 + 16(a^2 - b^2) = 16a^2 \Rightarrow x = \frac{-4b \pm 4a}{8} \Rightarrow \frac{a-b}{2},\ -\frac{a+b}{2}$	1+1+1
	OR: $\frac{15}{x(15-x)} = \frac{3}{10} \Rightarrow x^2 - 15x + 50 = 0 \Rightarrow$ 5 and 10	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $x + 3y = 550,\ x + 5y = 850$ (ii) $y =$ ₹150, $x =$ ₹100 (iii) ₹1300 (iv) $100 + 150h = 700 \Rightarrow h = 4$ — yes	1+2+1+1
	OR: Meet at $(3, 2)$; y-intercepts $(0, 8), (0, -1)$; area $= \frac{1}{2} \times 9 \times 3 = 13.5$ sq units	3+2
20	(i) $x(x+5) = 300$ (ii) $(x+20)(x-15) = 0 \Rightarrow$ 15 rows, 20 chairs (iii) $x^2 + 5x - 350 = 0$, $D = 1425$ not a perfect square $\Rightarrow$ no whole-number solution $\Rightarrow$ not possible	1+2+2
	OR: $\frac{300}{x} - \frac{300}{x+10} = 1 \Rightarrow x^2 + 10x - 3000 = 0 \Rightarrow (x+60)(x-50) = 0 \Rightarrow$ 50 km/h; $6 - 5 = 1$ ✓	2+2+1

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ANSWER KEY & MARKING SCHEME SET D

Chapters 3 & 4: **Pair of Linear Equations in Two Variables, Quadratic Equations** | Class 10 Mathematics | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** coincident	All ratios $\frac{1}{2}$
2	**(a)** $(2, 2)$	$x = y = 2$
3	**(c)** $(\frac{3}{2}, 4)$	Ratios $= \frac{1}{2}$
4	**(c)** Incorrect — it has infinitely many solutions	Every point on the line
5	**(b)** Student A is correct	Coefficients follow the quantities
6	**(c)** ± 3	$x^2 = 9$
7	**(b)** $-\frac{9}{4}$	$\frac{3}{4} - k - 3 = 0$
8	**(d)** 1	$49 - 48$
9	**(c)** Incorrect — -6 and -5 also work	$x^2 + x - 30 = 0 \Rightarrow x = 5$ or -6
10	**(a)** $(x + 2)^2 = x^2 + 3$	Reduces to $4x + 1 = 0$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$s = t + 3 \Rightarrow 2t + 6 + 3t = 36 \Rightarrow t = 6,\ s = 9$	1+1
12	Coincident ⇒ consistent (dependent); e.g. $(1, 4), (2, 3)$	1+1
13	$(x - a)^2 = b^2 \Rightarrow x = a + b$ or $a - b$	1+1
14	$D = 33 \Rightarrow x = \frac{-1 \pm \sqrt{33}}{4}$	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$a - b = 6,\ a + 3 = 2(b - 3) \Rightarrow b = 15,\ a = 21$	1+1+1
16	$x(x + 4) = 96 \Rightarrow (x + 12)(x - 8) = 0 \Rightarrow$ 8 and 12 (or -12 and -8)	1+1+1
17	Meet at $(3, 2)$; y-intercepts $(0, 11)$ and $(0, -1)$; vertices of region $(3, 2), (0, 11), (0, -1)$	2+1
	OR: $5x - 4y = -1,\ 2x - y = 5 \Rightarrow x = 7,\ y = 9 \Rightarrow \frac{7}{9}$	1+1+1
18	$x + y = 35,\ x^2 + y^2 = 625 \Rightarrow x^2 - 35x + 300 = 0 \Rightarrow$ 15 cm and 20 cm	1+1+1
	OR: $x^2 - 6x + 8 = 0$ — quadratic; $(x - 2)(x - 4) = 0 \Rightarrow$ 2, 4	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $x + 40y = 14000,\ x + 60y = 16000$ (ii) $y =$ ₹100, $x =$ ₹10000 (iii) ₹17,500 (iv) 100 items	1+2+1+1
	OR: $x + y = 25,\ 3x - y = 15$ (1); graph: lines meet at $(10, 15)$ (3); pen ₹10, box ₹15 — check $40 + 60 = 100,\ 30 = 15 + 15$ ✓ (1)	1+3+1
20	(i) $\frac{4800}{x-4} - \frac{4800}{x} = 40 \Rightarrow x^2 - 4x - 480 = 0$ (ii) $(x - 24)(x + 20) = 0 \Rightarrow$ 24 (iii) ₹200	2+2+1
	OR: (i) $x^2 + 3x - 180 = 0 \Rightarrow (x + 15)(x - 12) = 0 \Rightarrow$ 12 m × 15 m (ii) perimeter 54 m ⇒ ₹1080 (iii) Wrong — 15 m is the length	3+1+1

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